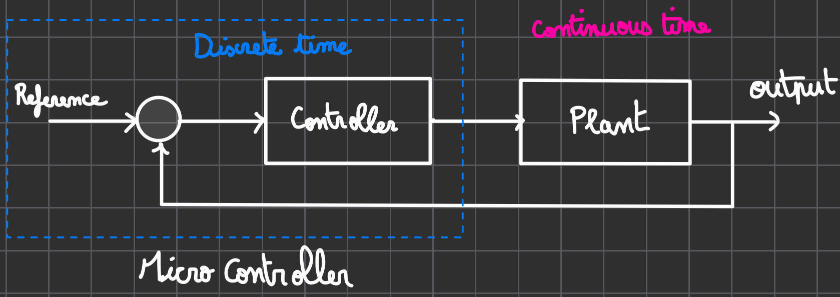
04) Discrete-Time System



	Continuous time		Discrete time
Transfert function	$G(s) = \dfrac{2s+5}{s^2+3s+2}$	$\Leftrightarrow$	$G(z) = \dfrac{z+0{,}9}{z^2+0{,}9z+0{,}7}$
	$\Updownarrow$		$\Updownarrow$
State Equation	$\dot{x}(t) = A\,x(t) + B\,u(t)$ $y(t) = C\,x(t)$	$\Leftrightarrow$	$x[k+1] = A\,x[k] + B\,u[k]$ $y[k] = C\,x[k]$

Geometric progression a_n Common ratio r

$$a_{n+1} = a_n\, r \quad \overset{\text{Similar}}{\Longleftrightarrow} \quad x(k+1) = A\,x(k)$$

$$\Downarrow$$

$$a_n = a_0\, r^{n-1}$$

$$\boxed{\begin{array}{l} \text{if } -1 < r < 1 \\ \lim\limits_{n\to\infty} a_n = 0 \\ \text{Converge to zero.} \end{array}}$$

otherwise a_n diverge

State feedback:

$$x(k+1) = a\,x(k) + b\,u(k)$$

$$a = 1{,}1 \quad , \quad b = 1 \quad \Longrightarrow \quad x(k+1) = 1{,}1\,x(k) + u(k)$$

$\notin \,]-1, 1[$ so we have to pay attention to the value of $u(k)$!!!

if $u(k) = 0$: diverge $\longrightarrow x(k+1) = 1,1\, x(k)$

if $u(k) = 1$: diverge $\longrightarrow x(k+1) = 1,1\, x(k) + 1$

if $u(k) = -0,2\, x(k)$: stable $\longrightarrow x(k+1) = 1,1\, x(k) - 0,2\, x(k)$

↳ Stabilized by state feedback

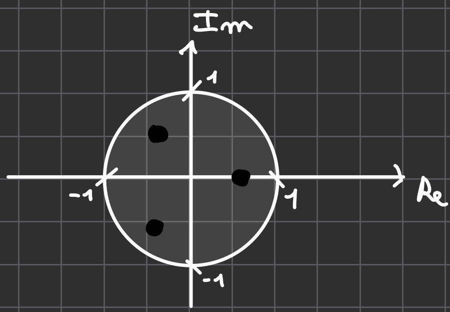
$\Leftrightarrow x(k+1) = x(k)\,[1,1 - 0,2]$

$\Leftrightarrow x(k+1) = \boxed{0,9}\, x(k)$

$\in]-1, 1[$

Stability in Control.

For a discrete-time system, the necessary and sufficient condition for asymptotic stability is that all eigenvalues of the matrix A lie within the unit circle.



Unit circle

Time variation of Lyapunov fonction values

$V = x^T(k)\, P\, x(k) \qquad P > 0$

Difference of Lyapunov function:

$$V(k+1) - V(k) = x^T(k+1)\, P\, x(k+1) - x^T(k)\, P\, x(k)$$

$$= x^T(k)\,(A^T P A - P)\, x(k)$$

explanations:

$x(k+1) = A\,x(k)$

substitution

$x^T(k+1)\, P\, x(k+1) \Longrightarrow (A\,x(k))^T\, P\, (A\,x(k))$

Transposition's rule

$x^T(k)\, A^T P A\, x(k) - x^T(k)\, P\, x(k)$

factorisation

$x^T(k)\,[A^T P A - P]\, x(k)$

Lyapunov Equation: $A^T P A - P + Q = 0 \Leftrightarrow Q = -A^T P A + P$

$$\Delta V = x^T(k)\left[A^T P A - P\right] x(k) \quad < \quad -x^T(k)\, Q\, x(k)$$

$$Q > 0$$

For any x, $V > 0$ and $\Delta V < 0$ hold $\Rightarrow$ V goes to zero.

↳ System is stable.

Example:

$$x(k+1) = 0{,}9\; x(k)$$

Lyapunov function:

$$V(k) = p\, x(k)^2 \quad , \quad p = 1$$

Time variation of Lyapunov function:

$$\underbrace{V(k+1)}_{\substack{=(p\,x(k+1))^2,\; p=1 \\ =(0{,}9\,x(k))^2}} - V(k) = 0{,}81\, x(k)^2 - x(k)^2 = \underline{-0{,}19\, x(k)^2}$$

↳ this value is smaller than zero for any x $\longrightarrow$ V is monotonically reduced.

$$q = 0{,}19 \;,\; a^2 p - p + q = 0$$

$$\Rightarrow 0{,}9^2 \times 1 - 1 + q = 0$$

$$\Leftrightarrow \quad q = -0{,}81 + 1$$

2nd order system:

$$x(k+1) = A\, x(k)$$

$$A = \begin{bmatrix} 0{,}9 & 0{,}2 \\ -0{,}2 & 0{,}9 \end{bmatrix}$$

Lyapunov equation:

$$A^T P A - P + Q = 0$$

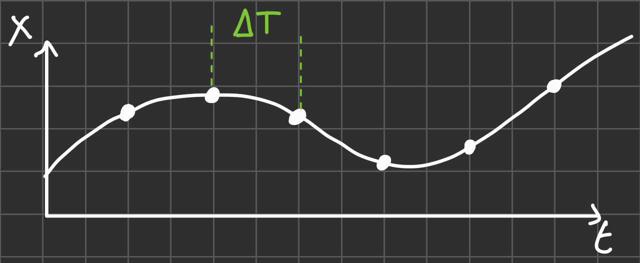
$$V(k) = x^T(k)\, P\, x(k)$$

$$\begin{aligned}
\Delta V &= x^T(k+1)\, P\, x(k+1) - x^T(k)\, P\, x(k) \\
&= (A\,x(k))^T P A\, x(k) - x^T(k) P\, x(k) \\
&= x^T(k)\, A^T P A\, x(k) - x^T(k) P\, x(k) \\
&= x^T(k) \left(A^T P A - P\right) x(k)
\end{aligned}$$

With $P = \begin{pmatrix} 6{,}6667 & 0 \\ 0 & 6{,}6667 \end{pmatrix}$ and $Q = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

(cf. Matlab simulation "LyapunovExample.m")

Continious Time to Discrete Time.



$$\frac{x(k+1) - x(k)}{\Delta T} = A\,x(k) + B\,u(k)$$

if ΔT is sufficiently small, almost same dynamics can be expressed by the discrete time system expression.

Method 2:

State after ΔT is given by following equation.

$$\boxed{x(t+\Delta T) = e^{A\Delta T} x(t) + \int_t^{t+\Delta T} e^{A(t-\tau)} B\, u(\tau)\, d\tau .}$$

$\downarrow$ Input $u(t)$ is fixed in the range $[t, t+\Delta T]$

$$x(t+\Delta T) = \underbrace{e^{A\Delta T}}_{A} x(t) + \Big(\underbrace{\int_t^{t+\Delta T} e^{A(t+\tau)} B\, d\tau}_{B} \Big)\, u(t)$$